Precept 7: Probability Exercises

October 27, 2026

Problem

The table below gives $P(x, y)$.

		x = 1	2	3
	1	0.02	0.06	0.04
y	2	0.10	0.36	0.22
	3	0.04	0.10	0.06

1. Verify this is a proper joint distribution.
2. What is the marginal $P(x)$?
3. What is the marginal $P(y)$?
4. What is the conditional $P(y \mid x = 1)$?
5. What is the conditional $P(x \mid y = 3)$?
6. Compute $\mathbb{E}[x] = \sum_{x=1}^{3} x\, P(x)$. What is this?
7. Compute $\mathbb{E}\big[(x - \mathbb{E}[x])^2\big]$. What is this?
8. Show that $\mathbb{E}\big[(x - \mathbb{E}[x])^2\big] = \mathbb{E}\big[x^2\big] - \mathbb{E}\big[x\big]^2$

Solution

Question 1

To get a proper joint distribution, we need two things:

1. All entries must be non-negative,
2. All entries must sum to 1.

Indeed, all entries are non-negative. To show they sum to one, we can write

$$\underbrace{(0.02 + 0.06 + 0.04)}_{0.12} + \underbrace{(0.10 + 0.36 + 0.22)}_{0.68} + \underbrace{(0.04 + 0.10 + 0.06)}_{0.20} = 1$$

Question 2

To get the marginal $P(x)$, we sum out y, meaning that we sum down each column.

$$P(x = 1) = 0.02 + 0.10 + 0.04 = 0.16$$

$$P(x = 2) = 0.06 + 0.36 + 0.10 = 0.52$$

$$P(x = 3) = 0.04 + 0.22 + 0.06 = 0.32$$

As a sanity check, $0.16 + 0.52 + 0.32 = 1$.

Question 3

To get the marginal $P(y)$, we sum out x by summing across each row.

$$P(y = 1) = 0.02 + 0.06 + 0.04 = 0.12$$

$$P(x = 2) = 0.10 + 0.36 + 0.22 = 0.68$$

$$P(x = 3) = 0.04 + 0.10 + 0.06 = 0.20$$

As a sanity check, $0.12 + 0.68 + 0.20 = 1$.

Question 4

The conditional distribution is

$$P(y \mid x = 1) = \frac{P(x = 1, y)}{P(x = 1)}$$

Using $P(x = 1) = 0.16$, we can write

$$P(y = 1 \mid x = 1) = \frac{0.02}{0.16} = 0.125, \quad P(y = 2 \mid x = 1) = \frac{0.10}{0.16} = 0.625, \quad P(y = 3 \mid x = 1) = \frac{0.04}{0.16} = 0.25$$

which sums to 1, as every distribution should.

Question 5

Similarly to **Question 4**, we grab the $y = 3$ row and divide by $P(y = 3) = 0.20$ to get

$$P(x = 1 \mid y = 3) = \frac{0.04}{0.20} = 0.2, \quad P(x = 2 \mid y = 3) = \frac{0.10}{0.20} = 0.5, \quad P(x = 3 \mid y = 3) = \frac{0.06}{0.20} = 0.3$$

Question 6

$$\mathbb{E}[x] = 1(0.16) + 2(0.52) + 3(0.32) = 0.16 + 1.04 + 0.96 = 2.16$$

This is the mean!

Question 7

$$\begin{aligned}
\mathbb{E}\big[(x - \mathbb{E}[x])^2\big] &= \sum_{x=1}^{3} (x - 2.16)^2 \, P(x) \\
&= (-1.16)^2(0.16) + (-0.16)^2(0.52) + (0.84)^2(0.32) \\
&= 1.3456(0.16) + 0.0256(0.52) + 0.7056(0.32) \\
&= 0.2153 + 0.0133 + 0.2258 \\
&= 0.4544
\end{aligned}$$

This is the variance!

Question 8

The trick is that the expectation function is linear! Hence, we can do the following:

$$\begin{aligned}
\mathbb{E}\big[(x - \mathbb{E}[x])^2\big] &= \mathbb{E}\big[x^2 - 2x\mathbb{E}\big[x\big] + \mathbb{E}\big[x\big]^2\big] \\
&= \mathbb{E}\big[x^2\big] - \mathbb{E}\big[2x\mathbb{E}\big[x\big]\big] + \mathbb{E}\big[\mathbb{E}\big[x\big]^2\big] \\
&= \mathbb{E}\big[x^2\big] - 2\mathbb{E}\big[x\big]\mathbb{E}\big[x\big] + \mathbb{E}\big[x\big]^2 \\
&= \mathbb{E}\big[x^2\big] - 2\mathbb{E}\big[x\big]^2 + \mathbb{E}\big[x\big]^2 \\
&= \mathbb{E}\big[x^2\big] - \mathbb{E}\big[x\big]^2
\end{aligned}$$